

Notes: Fisher, Gavazza, Liu, Ramadorai & Tripathy (JFE, 2024) Refinancing Cross-Subsidies in the Mortgage Market

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1 Big picture

- **Question.** How large are the implicit cross-subsidies created by mortgage refinancing frictions, and who pays them?
- **Setting.** The paper studies the UK mortgage market, where the dominant mortgage is a dual-rate contract: borrowers initially pay a low discounted rate, then automatically roll onto a higher reset rate unless they refinance.
- **Main mechanism.** Active households refinance back to the low discounted rate. Inactive households remain on higher reset rates. The extra revenue from inactive households permits low discounted rates for active households.
- **Core measurement problem.** Cross-subsidies are not directly observed because they depend on the counterfactual contract that would have prevailed without the dual-rate structure.
- **Method.** The authors build and estimate a structural model of mortgage demand and refinancing. They then compare the observed dual-rate market with a counterfactual single-rate market with no refinancing requirement.
- **Data.** The model is estimated using UK Financial Conduct Authority administrative data covering the stock of outstanding UK mortgages at semiannual frequency, with the main exercise focused on 2015H1.
- **Main empirical fact.** Lower-income households are less likely to be on discounted rates and are more likely to pay reset rates. Higher-income households have larger mortgages and are more likely to refinance.
- **Estimated refinancing costs.** The estimated average refinancing cost is roughly £4,042 among mortgage borrowers, with a large standard deviation of about £15,102.
- **Main counterfactual.** Replacing the dual-rate system with a constant single rate increases the number of mortgages, reduces average loan size, and raises consumer surplus.
- **Distributional result.** Cross-subsidies are regressive: poorer households and households in less wealthy regions implicitly subsidize richer households and households in wealthier areas.
- **Economic takeaway.** Product design in household finance can redistribute wealth even without explicit fees or taxes. Complexity and required household action can generate regressive transfers.

2 Data and measurement

2.1 Dual-rate mortgage contracts

- UK mortgages commonly feature a discounted rate for an initial fixed period.
- When the initial period expires, the borrower automatically rolls onto a higher reset rate unless they refinance.
- Refinancing is valuable when the interest-rate saving is larger than the household's refinancing cost.
- Refinancing costs include monetary fees, time costs, psychological costs, attention costs, search costs, and inertia.

Interpretation: The contract is attractive to active borrowers but costly for inactive borrowers. The key economic question is whether the higher reset-rate payments from inactive borrowers subsidize active borrowers who obtain low discounted rates.

2.2 Administrative mortgage data

- The paper uses rich administrative data from the UK Financial Conduct Authority.

- The data cover mortgages issued by regulated financial institutions and track the stock of outstanding mortgages at semiannual frequency from 2015H1 to 2017H2.
- The main quantitative analysis focuses on the 2015H1 mortgage stock.
- After filters, Table 1 reports 3,590,228 mortgage observations.
- The sample distinguishes mortgages paying the discounted rate from those paying the reset rate.

Interpretation: The comprehensive stock data let the authors observe how many borrowers remain on reset rates, how large those loans are, and how these patterns vary with income and geography.

2.3 Key mortgage variables

- **Current loan balance:** outstanding mortgage balance.
- **Interest rate:** rate currently paid, separately for discounted-rate and reset-rate mortgages.
- **Original loan balance and term:** mortgage size and contract maturity at origination.
- **Remaining term and discount period:** how long remains on the mortgage and discounted period.
- **Income quantiles:** the main observable grouping used to study distributional effects.

Interpretation: The loan balance matters because refinancing benefits scale with outstanding debt. A large mortgage makes it worthwhile to pay fixed refinancing costs.

2.4 Baseline descriptive statistics

- In 2015H1, the mortgage stock is split between discounted-rate mortgages and reset-rate mortgages.
- Discounted-rate mortgages have an average current loan balance of about £140,647.
- Reset-rate mortgages have an average current loan balance of about £112,692.
- Discounted-rate mortgages pay an average annual interest rate of 3.20 percent.
- Reset-rate mortgages pay an average annual interest rate of 3.72 percent.
- Roughly 65 percent of mortgages in the sample pay the discounted rate.

Interpretation: The interest-rate gap is economically meaningful. It is the raw source of potential gains from refinancing and the raw source of cross-subsidies.

2.5 Income variation

- Homeownership rises sharply with income: from 50 percent in the lowest income group to 96 percent in the highest group.
- Mortgage balances also rise sharply with income: from about £61,522 in the lowest group to about £351,530 in the top group.
- The share of borrowers on discounted rates rises with income.
- Higher-income borrowers therefore both have more to gain from refinancing and appear more likely to refinance.

Interpretation: Income is the key dimension for studying regressivity. If lower-income borrowers are more likely to pay reset rates, the dual-rate contract can shift surplus away from poorer households.

3 Model and empirical specifications

3.1 Household choice and mortgage payment

A measure M of households enters in each period. Household i chooses whether to rent or buy a house with a mortgage. If the household buys, it obtains flow utility

$$v_i h_i^\alpha - m(l_i, r, T), \quad (1)$$

where v_i is the valuation of housing, h_i is house size, l_i is loan balance, r is the mortgage rate, and T is mortgage maturity.

The per-period amortizing mortgage payment is

$$m(l_i, r, s) = l_i \frac{r(1+r)^s}{(1+r)^s - 1}. \quad (2)$$

Interpretation: The housing-valuation parameter explains why households choose different loan sizes. The payment formula links interest rates to mortgage demand and refinancing incentives.

3.2 Loan balance dynamics

The outstanding loan balance evolves according to

$$l_{i,t+1}(r, l_{i,t}) = l_{i,t}(1+r) - m(l_{i,t}, r, s). \quad (3)$$

Interpretation: Loan amortization means that refinancing benefits decline over time. A borrower with a small remaining balance may not refinance even if the reset rate is high.

3.3 Refinancing costs

A household can refinance back to the discounted rate r_d by paying a refinancing cost

$$k_{i,t} = k_i \varepsilon_{i,t}, \quad (4)$$

where k_i is a persistent household-specific component and $\varepsilon_{i,t}$ is a transitory shock.

Interpretation: This captures persistent differences in sophistication or attention, while allowing occasional refinancing when a normally inactive household receives a low temporary cost shock.

3.4 Refinancing threshold

In the last refinancing period, a household refinances if its cost is below the interest saving:

$$k_i^*(T) = m(l_i, R, 1) - m(l_i, r_d, 1) = l_i(R - r_d), \quad (5)$$

where R is the reset rate.

Interpretation: Larger loans create larger monetary benefits from refinancing. This is why high-income borrowers with larger balances are more likely to refinance.

3.5 Expected payment value function

The model recursively computes the expected payment burden for a borrower facing future refinancing decisions. In the final refinancing period, the value is

$$V_T(k_i, l_i) = \int_0^{k_i^*(T)/k_i} [m(l_i, r_d, 1) + k_i \varepsilon] dF(\varepsilon) + \int_{k_i^*(T)/k_i}^\infty m(l_i, R, 1) dF(\varepsilon). \quad (6)$$

Interpretation: The first integral covers states in which the borrower refinances. The second covers states in which the borrower remains on the reset rate.

3.6 Single-rate counterfactual

The main counterfactual replaces the dual-rate mortgage with a single constant rate and no refinancing requirement. The benchmark counterfactual rate equals the weighted average rate in the data:

$$r_c = \frac{r_d (Q_0(r_d) + Q_1(r_d)) + RQ_2(R)}{Q_0(r_d) + Q_1(r_d) + Q_2(R)}. \quad (7)$$

Interpretation: The single-rate economy removes the need for costly household action. It also removes the discounted-rate advantage of active refinancers.

3.7 Minimum-distance estimation

The estimated parameter vector ψ minimizes

$$[m(\psi) - m^S]' \Omega [m(\psi) - m^S], \quad (8)$$

where $m(\psi)$ is the model-implied moment vector, m^S is the sample moment vector, and Ω is the optimal weighting matrix.

Interpretation: The model is disciplined by matching both aggregate mortgage moments and moments that vary across borrowers. This allows the authors to infer unobserved refinancing costs.

4 Identification and empirical design

4.1 What is identified by the data

- The distribution of loan balances identifies heterogeneity in housing valuations.
- The distribution of discounted-rate and reset-rate mortgages identifies refinancing behavior.
- The transition from reset to discounted rates helps identify refinancing-cost heterogeneity.
- Cross-income differences in mortgage balances and discounted-rate shares discipline the distributional counterfactuals.

Interpretation: The paper does not estimate a reduced-form causal treatment effect. It uses a structural model to infer the unobserved costs that rationalize observed mortgage choices.

4.2 Why a structural counterfactual is needed

- Refinancing behavior depends on both financial benefits and unobserved costs.
- Loan demand changes when contract structure changes.
- Participation in homeownership and mortgage borrowing changes in the single-rate counterfactual.
- Cross-subsidies require comparing the dual-rate system with a non-observed alternative system.

Interpretation: A reduced-form comparison cannot separately identify financial incentives, refinancing frictions, and counterfactual demand responses.

4.3 Model credibility checks

- The model matches key data moments on loan balances, remaining maturities, discounted-rate shares, and transition rates.
- It reproduces the pattern that larger loan balances have higher probabilities of transitioning from reset to discounted rates.
- The model is estimated both UK-wide and separately across income groups and regions.
- The counterfactual is evaluated under a transparent benchmark interest rate equal to the weighted average rate in the observed system.

Interpretation: The main credibility comes from fitting multiple moments that are informative about refinancing incentives and costs.

4.4 Limitations

- The model assumes steady state.
- Discounted and reset rates are treated as exogenous in the baseline counterfactual.
- Supply-side lender responses are not separately modeled.
- The model abstracts from cash-out refinancing and maturity extension.
- The monetary measure of refinancing costs includes psychological, attention, and inertia costs, not only direct fees.

Interpretation: The findings should be read as a disciplined structural accounting of cross-subsidies, not as a randomized policy experiment.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format, with adjacent figures and tables grouped together to save space.

5.1 Tables 1–2: mortgage stock and income groups

Read:

- Table 1 reports summary statistics for 3.59 million mortgages in 2015H1.
- The average discounted-rate mortgage balance is £140,647; the average reset-rate mortgage balance is £112,692.
- Discounted-rate mortgages pay 3.20 percent on average; reset-rate mortgages pay 3.72 percent.
- Table 2 shows that homeownership, loan balance, and discounted-rate use all rise with income.
- The bottom income group has a 50 percent homeownership rate and average balance of £61,522; the top group has a 96 percent homeownership rate and average balance of £351,530.

Economic message: The descriptive statistics already point toward regressive cross-subsidies: lower-income households are less likely to be on the discounted rate, while higher-income households have larger loans and more to gain from refinancing.

Table 1
Summary statistics for the mortgage stock in 2015H1.

	Mean	SD	Median
Current Loan Balance, Discounted Rate (£)	140,647	105,062	114,953
Current Loan Balance, Reset Rate (£)	112,692	79,684	93,916
Interest Rate, Discounted (%)	3.20	0.95	3.14
Interest Rate, Reset (%)	3.72	0.98	3.99
Original Loan Balance (£)	142,333	100,661	118,399
Original Term (Years)	23.32	7.07	25.00
Remaining Term, Discounted Rate (Years)	20.57	7.73	20.92
Remaining Term, Reset Rate (Years)	16.84	6.95	16.50
Remaining Discounted Period (Years)	2.11	1.52	1.83
Borrower Age (Years)	41.97	10.02	41.00
Observations	3,590,228		

Notes: The table above shows summary statistics of mortgages from the stock data reported in 2015H1. The sample includes mortgages in two categories, namely, those paying discounted interest rates, and those paying the reset rate ("Standard Variable Rate"). The total sample comprises around 3.59 million mortgages, of which 65 percent are discounted rate mortgages at this point in time. Table A.1 in Appendix A contains a description of the underlying variables.

Table 1: mortgage stock in 2015H1.

Summary statistics for the mortgage stock in 2015H1, by income quantiles.

Quantiles	Income (£)	Homeowners (%)	Balance (£)	Discounted (%)
0-10	24,604	0.50	61,522	0.65
10-20	30,504	0.61	76,141	0.64
20-30	35,631	0.64	86,894	0.64
30-40	40,701	0.68	96,752	0.64
40-50	46,198	0.72	107,326	0.64
50-60	52,687	0.75	118,920	0.64
60-70	60,974	0.80	132,885	0.65
70-80	73,154	0.82	152,206	0.66
80-85	82,349	0.84	173,235	0.66
85-90	96,616	0.86	195,018	0.67
90-95	126,236	0.91	232,354	0.68
95-100	214,486	0.96	351,530	0.69

Notes: The table above shows summary statistics of mortgages from the stock data reported in 2015H1, split by income quantiles of borrowers. Table A.1 in Appendix A contains a description of the underlying variables.

Table 2: mortgage stock by income quantile.

5.2 Tables 3–4: estimated parameters and model fit

Read:

- Table 3 reports the model parameters estimated UK-wide and for income-group and regional versions.
- The two-year discounted and reset rates are 650 bps and 759 bps.
- The UK-wide estimates imply a mixture distribution of origination/refinancing costs and substantial heterogeneity in refinancing costs.
- Table 4 shows that the model broadly matches key moments, including loan balances, remaining terms, discounted-rate shares, reset-to-discount transition rates, and ownership shares.

Economic message: The model's parameters transform observed refinancing choices into estimates of unobserved refinancing costs. The model fit is essential because the counterfactual subsidies depend on the estimated distribution of those costs.

Table 3
Parameters.

	r	R	ω	T	η
	650	759	1.2500	15	0.5000
β	0.9025				
	UK-wide (1)	Income Groups (2)	Regions (3)		
μ_c	0.0008 (0.0000)	0.0007 (0.0002)	0.0009 (0.0006)		
σ_c	0.1566 (0.0002)	0.2193 (0.0119)	0.3427 (0.0268)		
μ_{k1}	5.7454 (0.0076)	5.1239 (0.7421)	5.1073 (0.5263)		
σ_{k1}	2.6460 (0.0304)	1.7335 (0.2466)	1.5336 (0.9701)		
μ_{k2}	9.1875 (0.0178)	9.3004 (0.1201)	9.1938 (0.0643)		
σ_{k2}	0.9868 (0.0002)	0.9547 (0.0794)	0.9754 (0.0969)		
σ_e	0.9253 (0.0010)	0.8995 (0.0470)	0.8104 (0.0690)		
α	0.7856 (0.0000)	0.7896 (0.0076)	0.7879 (0.0026)		
$\bar{u}$	1062 (4.0946)	1,564 (590.5706)	1556 (430.2231)		
M	379,145	28,164 (11,890)	31,894 (15,298)		

Notes: This table reports the estimated parameters. In column (1), the numbers in parentheses refer to asymptotic standard errors of the parameter estimates. In columns (2) and (3) the numbers in parentheses

Table 3: parameters.

Table 4
Model fit.

	Data	Model	t-statistic
Mean Loan Balance, Discounted Rate	140,647	141,240	-8.63
Standard Deviation Loan Balance, Discounted Rate	105,062	107,233	-11.49
Mean Loan Balance, Reset Rate	112,692	111,688	14.12
Standard Deviation Loan Balance, Reset Rate	79,684	78,120	6.65
Mean Remaining Years, Discounted Rate	20.57	18.82	28.96
Standard Deviation Remaining Years, Discounted Rate	7.73	7.85	-3.30
Mean Remaining Years, Reset Rate	16.84	15.51	17.95
Standard Deviation Remaining Years, Reset Rate	6.95	7.39	-8.67
Share of Mortgages on Discounted Rate, 0-5 Percentile	52.72	52.88	-1.39
Share of Mortgages on Discounted Rate, 5-25 Percentile	56.36	57.62	-21.53
Share of Mortgages on Discounted Rate, 25-50 Percentile	61.48	59.95	29.80
Share of Mortgages on Discounted Rate, 50-75 Percentile	67.76	63.97	76.82
Share of Mortgages on Discounted Rate, 75-95 Percentile	73.77	71.88	36.55
Share of Mortgages on Discounted Rate, 95-100 Percentile	81.19	81.93	-8.02
Transition from Reset Rate to Discounted Rate	16.52	18.01	-45.10
Share of Owners	63.13	63.12	0.97

Notes: This table reports the values of the empirical moments and of the moments calculated at the estimated parameters reported in column (1) of Table 3.

Table 4: model fit.

5.3 Figures 1–2: refinancing benefits and transition probabilities

Read:

- Figure 1 left panel shows the distribution of net benefits from refinancing.
- The median net benefit is positive, but the average net benefit is negative because of a long left tail.

- Figure 1 right panel shows that some borrowers rarely refinance while others remain on discounted rates for many periods.
- Figure 2 shows that transition probabilities from reset to discounted rates rise with loan balance deciles.
- The model reproduces the positive relationship between loan size and refinancing.

Economic message: Refinancing is not purely a rate-arbitrage decision. It depends on heterogeneous costs, and large-balance borrowers are more likely to find refinancing worthwhile.

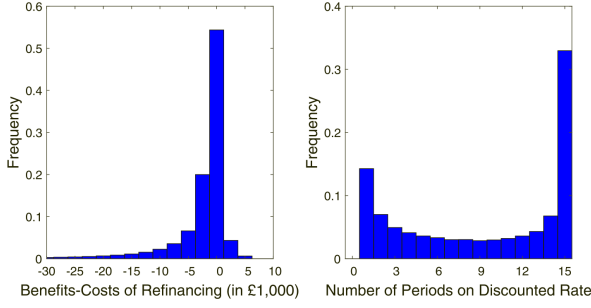


Figure 1: distribution of net benefits and discounted-rate periods.



Figure 2: transition probabilities by loan balance.

5.4 Figure 3, Table 5, and Figure 4: refinancing costs and single-rate counterfactual

Read:

- Figure 3 shows that estimated average refinancing costs rise with income.
- This does not contradict more frequent refinancing by high-income borrowers, because they also have much larger loans and larger refinancing benefits.
- Table 5 reports the main counterfactual with a constant interest rate of 683 bps.
- Under the single-rate counterfactual, the number of mortgages rises, average loan size falls, and consumer surplus rises.
- Figure 4 shows that the change in loan size is heterogeneous: some households reduce loan size, while high-cost borrowers can increase loan size because they no longer face repeated refinancing costs.

Economic message: Eliminating the dual-rate structure changes both the extensive margin and the intensive margin: more households borrow, but average initial loans become smaller.

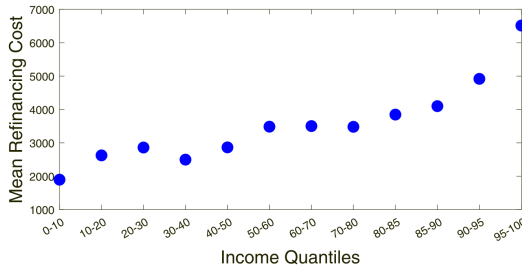


Figure 3: refinancing costs by income group.

Table 5

Market outcomes with constant interest rates.

	UK-wide	Income groups	Regions
Constant Interest Rate=683 bps			
Number of Mortgages	1.06	1.09	1.07
Mean Initial Loan Amount	0.97	0.97	0.97
Standard Deviation Initial Loan Amount	0.96	0.93	0.94
Mean Loan Balance	0.97	0.97	0.97
Standard Deviation Loan Balance	0.96	0.94	0.95
Consumer Surplus	1.04	1.03	1.03

Notes: This table reports the statistics on the mortgage market in counterfactual markets with constant interest rates, as ratios of those of the estimated market with dual interest rates. The statistics are calculated using a

Table 5: counterfactual market outcomes.

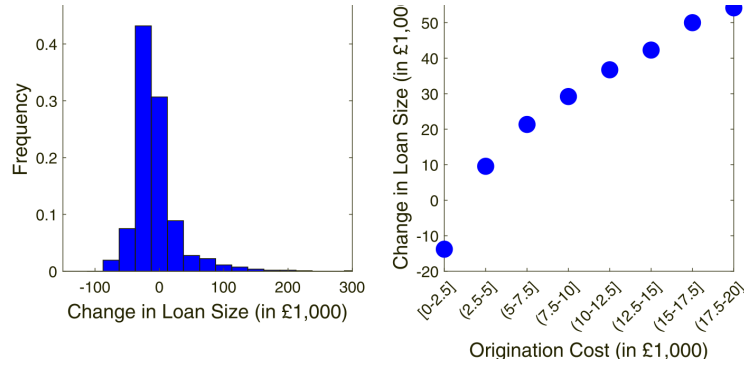


Figure 4: change in loan size at origination.

5.5 Figures 5–6: distributional effects across income groups and regions

Read:

- Figure 5 shows that moving to the single-rate economy lowers interest rates for lower-income groups and raises them for high-income groups.
- The share of households with mortgages increases most for lower-income groups.
- Average initial loan size falls more for high-income groups.
- Consumer surplus rises for all groups, but percentage increases are larger for lower-income groups.
- Figure 6 shows corresponding regional changes. Less wealthy regions tend to benefit more from the move away from dual-rate contracts.

Economic message: The current dual-rate mortgage structure generates regressive implicit transfers. Simplifying the contract reverses some of those transfers by benefiting poorer households and less wealthy regions.

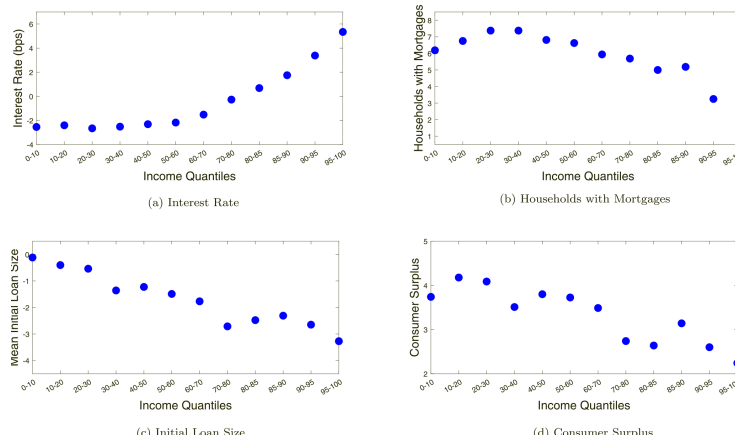


Figure 5: changes in market outcomes by income groups.

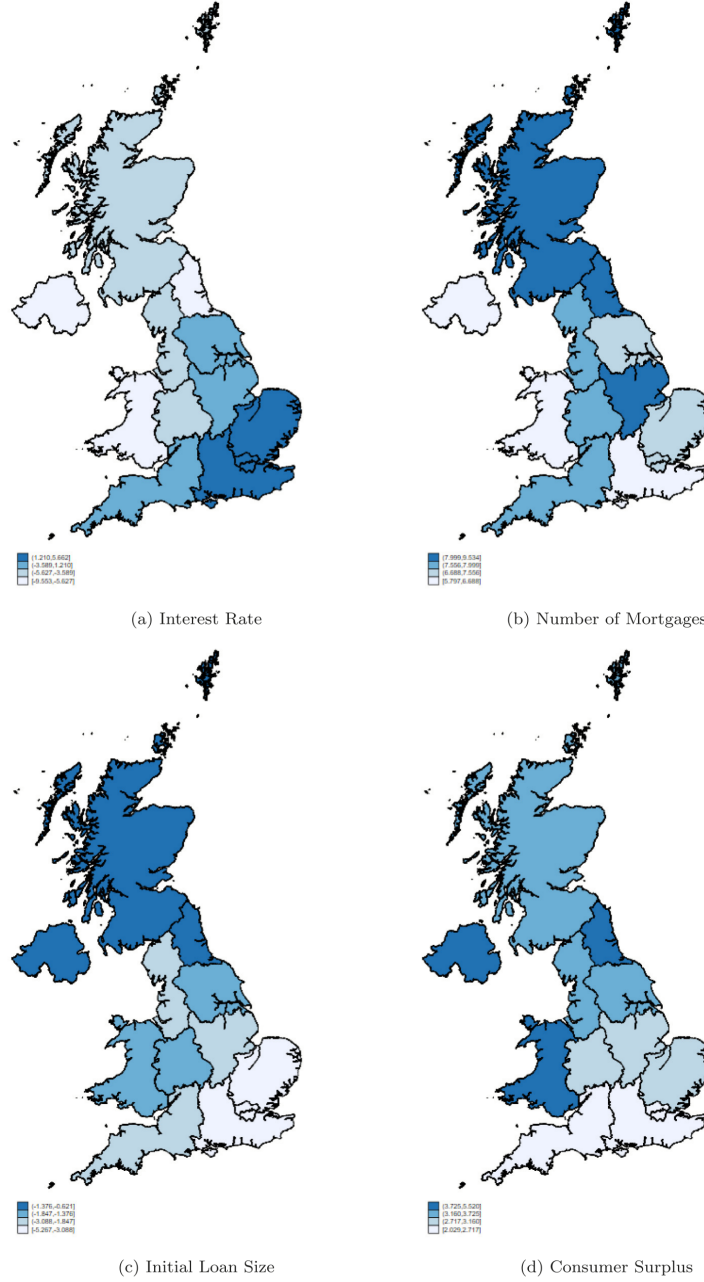


Figure 6: regional changes.

6 Short summary

- The paper studies hidden cross-subsidies created by dual-rate mortgage contracts in the UK.
- Borrowers initially pay a low discounted rate but roll onto a higher reset rate unless they refinance.
- Active borrowers refinance promptly; inactive borrowers continue paying high reset rates.
- The authors build a structural model of mortgage demand and refinancing to infer unobserved refinancing costs.

- The main data are UK FCA administrative mortgage-stock data, with the main exercise focused on 2015H1.
- The model explains heterogeneity in loan size, homeownership, discounted-rate use, and reset-to-discount transitions.
- Estimated refinancing costs are large and heterogeneous.
- The counterfactual single-rate economy removes the need to refinance.
- In that economy, the number of mortgages increases, average initial loan size decreases, and consumer surplus rises.
- Low-income households benefit more in percentage terms because they are more likely to pay reset rates in the observed system.
- High-income households lose some of the implicit subsidy they receive through discounted rates.
- The paper shows that household finance product design can contribute to wealth inequality.

7 Conclusion

7.1 Core conclusion

Fisher, Gavazza, Liu, Ramadorai, and Tripathy show that dual-rate mortgage contracts generate large hidden cross-subsidies. Households that fail to refinance pay high reset rates, and these payments subsidize low discounted rates for households that refinance quickly. Because poorer households are less likely to refinance and richer households have larger loans and refinance more often, the cross-subsidy is regressive.

7.2 Economic intuition

The central intuition is that contract complexity creates a transfer. If every borrower faced a transparent single rate, there would be no need to act at the reset date. In a dual-rate system, the borrower must act to avoid a high rate. Households with low refinancing costs or high benefits act; households with high costs or inattention do not. The resulting payments are not labeled as transfers, but they redistribute surplus across households.

7.3 Takeaway

The paper turns a familiar household finance friction—inattention to refinancing—into a measurable source of inequality. It shows that financial product design can have distributional consequences even when all contracts are voluntarily chosen and all prices are market prices.